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B.Tech / M.Tech (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

Third Semester

21CSC201J - DATA STRUCTURES AND ALGORITHMS

(For the candidates admitted from the academic year2021-2022 to 2023-2024)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours | Max. Marks: 75

PART - A (20 x 1 = 20 Marks)
Answer ALL Questions

	Question	Marks	BL	CO	PO
1.	______are the set of definitions that allow us to use the functions while hiding the implementation A) Abstract Data Types B) Algorithms C) Referential Structures D) Datatypes	1	1	1	2

2.	What is the output if it is executed on a 32 bit processor?
	void main()
	{
	int *p;
	p = (int *)malloc(20);
	printf("%d", sizeof(p));
	}
	A) 2 B) 4
	C) 8 D) Compiler dependent
		1	1	1	2

3.	Which of the following functions allocates multiple blocks of memory, each block of the same size? A) malloc() B) realloc() C) calloc() D) free()	1	1	1	3
4.	We can certainly deduce the ______ scenario of an algorithm from its asymptotic analysis. A) Best case B) Best, worst and average cases C) Worst case D) Average case	1	1	1	12
5.	Identify the correct option for creating a node in linked list at first time A) ptr = (struct node) malloc (sizeof(struct node *)); B) ptr = (struct node *)calloc(sizeof(struct node *)); C) ptr = (struct node *)malloc(sizeof(struct node)); D) ptr = (struct node *) calloc (size(struct node));	1	1	2	3
6.	The matrix contains m rows and n columns. The matrix is called Sparse Matrix if ________ A) Total number of Zero elements > (m*n)/2 B) Total number of Zero elements = m + n C) Total number of Zero elements = m/n D) Total number of Zero elements = m-n	1	1	2	3
7.	Find out the correct option when adding a new node into the end of list A) Head is made to point to the new node, New node points to the previously first element B) Last node now points to the new node, New node points to NULL C) Previous node now points to the new node, New node points to the next node D) Last node now points to the new node, New node points to the previously first element	1	2	2	2

8. Which of the following is a practical example of a doubly linked list 1 1 2 3

A) A browser cookie file
B) A quest in a game that lets users retry stages
C) A game in which the player runs forward
D) A first-in-first out scheduling system

9. A stack data structure cannot be used for 1 1 3 2

A) Implementation of Recursive Function
B) Allocation Resources and Scheduling
C) Reversing string
D) Evaluation of string in postfix form

10. The five following names Allen, Brindha, Chitra, Divya and Elon are pushed in a stack, one after another starting from Allen. The stack is popped four times and each name is inserted in a Queue. Then two names are deleted from the Queue and pushed back on the stack. Now one name is popped from stack. The popped name is ___________. 1 3 3 2

A) Allen
B) Brindha
C) Chitra
D) Divya

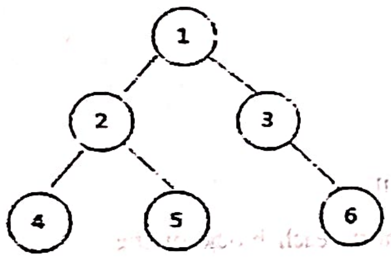
11. If the elements "A", "B", "C" and "D" are placed in a queue and are deleted one at a time, in what order will they be removed? 1 1 3 2

A) ABCD
B) DCBA
C) DCAB
D) ABDC

12. Identify the Postfix form of following expression. D + (E * F^G) 1 1 3 12

A) DFG^E * +
B) DEFG^* +
C) GFED^ *+
D) GF^E* D+

13. Which one is the correct In-order traversal for the given tree 1 1 4 3



A) 1->2->3->4->5->6
B) 4->2->5->1->3->6
C) 4->5->2->6->3->1
D) 4->2->5->3->6->1

14. A B-tree of order 4 and of height 3 will have a maximum of ______ keys. 1 1 4 2

A) 255
B) 63
C) 127
D) 188

15. What is a hash table? 1 1 4 3

A) A structure that maps values to keys
B) A structure used for storage
C) A structure that maps keys to values
D) A structure used to implement stack and queue

16. If several elements are competing for the same bucket in the hash table, what is it called? 1 1 4 3

A) Diffusion
B) Replication
C) Collision
D) Duplication

17. Which of the following statements is/are TRUE for an undirected graph? 1 1 5 12

P: The number of odd-degree vertices is even

Q: Sum of degrees of all vertices is even

A) P Only
B) Q only
C) Both P and Q
D) Neither P nor Q

18. Consider an undirected unweighted graph G. Let a breadth-first traversal of G be done starting from a node r. Let d(r, u) and d(r, v) be the lengths of the shortest paths from r to u and v respectively, in G. If u is visited before v during the breadth-first traversal, which of the following statements is correct? 1 1 5 12

A) d(r, u) < d (r, v)
B) d(r, u) > d(r, v)
C) d(r, u) <= d (r, v)
D) d(r, u) =d(r, v)

19. The efficiency of open addressing techniques 1 1 5 3

A) Depends on the size of the table and number of elements in the table
B) Depends on the size of the table
C) Depends on total number of elements in the table
D) Depends on the hashing technique

20. Which of the following algorithms is commonly used to find a minimum spanning tree in a connected undirected graph? 1 1 5 3

A) Dijkstra's algorithm
B) Kruskal's algorithm
C) Bellman-Ford algorithm
D) Floyal-Warshall algorithm

PART - B (5 x 8 = 40 Marks)
Answer ALL Questions

Marks BL CO PO

21 a. Calculate the time complexity for the following code using Tabular method. Specify the asymptotic time complexity (worst case) for the given algorithm. 8 1 1 2

```
void calculate (int arr[], int n)
{
int sum = 0;
for (int i = 0; i < n; i++)
{
for (int j = i; j < n; j++)
{
for (int k = 0; k < 10; k++)
{ sum += arr[j];
} } } printf("%d\n", sum); }
```

(OR)

b. Discuss the concept of self-referential structures in the C programming language. How are they typically used in the implementation of dynamic data structures like linked lists? Illustrate with an example, including code snippets, to demonstrate how a self-referential structure can be used to implement a basic linked list. 8 1 1 1

22 a. Write an algorithm that takes in a two-dimensional array and find the maximum of all the row sums. A row sum is simply the sum of each number on that row? What are the limitations of array data structure 8 3 2 1

(OR)

b. Consider a goods train, which initially has an engine as a header. Later compartments are linked with engines one by one in series. The compartments can be attached and detached with an engine or in between the compartments. Implement a data structure for the following operations in the goods train 1. Attaching a compartment in between the compartments 2. Detaching a compartment in between the compartments. 8 3 2 2

23 a. Write the pseudocode to convert the expression given below into its corresponding postfix expression and then evaluate it. 8 3 3 12

$10 + ((7 - 5) + 10)/2$

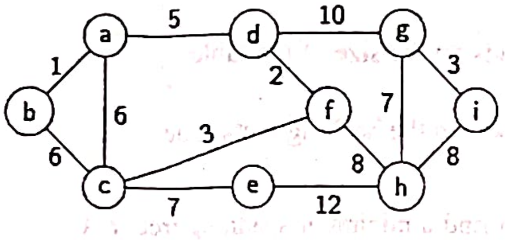
(OR)

b. Write the pseudocode to insert an element in the linked queue and delete an element from the linked queue with an example. 8 3 3 12

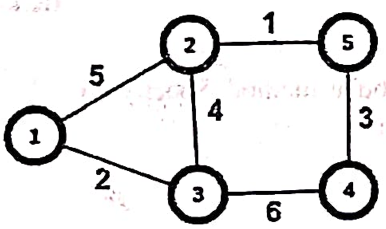
24 a. Mythili wants to constructs a tree for the given data **15, 10, 8, 12, 28, 20, 47, 37, 64 and 2** in such a way that the value of the nodes in right sub tree should be greater than the left sub tree. Then she felt that the leaves in the constructed tree will not be in same order, so she decided to remove the data 8, 28 and 64. Construct the tree as per Mythili's wish. 8 3 4 1

(OR)

b. Construct the minimum spanning tree using suitable algorithm and justify its efficiency 8 3 4 2



25 a. Draw the possible spanning trees for the given Graph G. Identify the minimum spanning tree among the spanning trees drawn. Write the pseudocode for generating spanning trees. 8 3 5 2



(OR)

b. Harish plans to have a small library at his home. He has lot of books and he wants to assign a unique number to each book. This unique number helps him in identifying the position of the books on the bookshelf. The size of the bookshelf is 10. Help him in place the books in the bookshelf without any collision. The entries are **69, 08, 39, 48, 15, 77, 19.** Compare the state of the structure by using two collision avoidance strategies. 8 3 5 2

PART - C (1 x 15 = 15 Marks)
Answer ANY ONE Question

Marks BL CO PO

26. A company approaches you to develop a customer support ticket system, they might add new tickets and remove tickets, from anywhere. Suggest a suitable abstarct data type. Implement using appropriate structure and provide pseudocode for 15 3 2 3

1. Insert ticket at the end
2. Remove ticket from the beginning
3. Insert at the beginning
4. Delete the last ticket

Justify why you suggested the ADT·

27. Construct two trees for the elements 11, 22, 33, 44, 55, 66, 77, 88 15 3 4 3

Property 1: Left subtree should have elements smaller than the element at parent and elements of right subtree should be greater than that of the parent.

Peroperty 2: Difference in the height of the left subtree and right subtree should not greater than 1.

First tree should satisfy property1 and the second tree should satisfy both the properties.

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